An example that a nef abundant divisor which is not log-abundant

We first recall some basic algebraic geometry. We work with algebraic schemes over an algebraically closed field k of characteristic zero, for instance $\mathbb{C}$.

Definition–Theorem 1 (Projective bundle)**.** Let S be a regular scheme and V be a rank-r vector bundle, that is, a locally free sheaf. The projective bundle

$$f : \mathbb{P}_S(V) = \mathrm{Proj}_S(\mathrm{Sym}^\bullet V) \longrightarrow S$$

is obtained by gluing $U \times \mathbb{P}^{r-1}$ over open subsets of S on which V is trivial, using changes of basis of V. There is a line bundle $\mathcal{O}_{\mathbb{P}_S(V)}(1)$ and a surjection

$$f^*V \twoheadrightarrow \mathcal{O}_{\mathbb{P}_S(V)}(1),$$

which we call the tautological quotient. A quotient $V \twoheadrightarrow L$ onto a line bundle defines a section $t : S \to \mathbb{P}_S(V)$ with $t^*\mathcal{O}_{\mathbb{P}_S(V)}(1) \cong L$. Here a section means $f \circ t = \mathrm{id}_S$.

Proof. See [Har77, II, 7.12 and Exercise 7.10]. □

Proposition 2. *Let S be a regular scheme, V a rank-r vector bundle, and $f : X = \mathbb{P}_S(V) \to S$ the projective bundle. Then*

(1) *For any integer $m \geq 0$ and line bundle L on S, we have*

$$f_*\big(\mathcal{O}_X(m) \otimes f^*L\big) \cong \mathrm{Sym}^m V \otimes L.$$

(2) *X is a smooth k-scheme, and*

$$\omega_X \cong \mathcal{O}_X(-r) \otimes f^*(\omega_S \otimes \det V).$$

Proof. There is an elementary fact that

$$H^0\big(\mathbb{P}^{r-1}_A, \mathcal{O}(m)\big) = A[T_1, \ldots, T_r]_m$$

for every $m \geq 0$. So the natural homomorphism $\mathrm{Sym}^m V \to f_*\mathcal{O}_X(m)$ is an isomorphism on affine open subsets trivializing V, hence patches together to an isomorphism. Then use the projection formula:

$$f_*\big(\mathcal{O}_X(m) \otimes f^*L\big) \cong \mathrm{Sym}^m V \otimes L.$$

This proves (1).

For (2), since X is locally of the form $U \times \mathbb{P}^{r-1}$, f is a smooth morphism. Since S is smooth, we see that X is smooth over $\mathrm{Spec}\, k$. The relative Euler sequence (see [Har77, III, Exercise 8.4(b)])

$$0 \longrightarrow \Omega_{X/S} \longrightarrow f^*V \otimes \mathcal{O}_X(-1) \longrightarrow \mathcal{O}_X \longrightarrow 0$$

gives that

$$\omega_{X/S} \cong f^*(\det V) \otimes \mathcal{O}_X(-r).$$

Then

$$\omega_X \cong f^*\omega_S \otimes \omega_{X/S} \cong \mathcal{O}_X(-r) \otimes f^*(\omega_S \otimes \det V).$$

□

Theorem 3 (Riemann–Roch formula)**.** *Let C be a smooth projective curve of genus g, and let L be a line bundle on C. Then*

$$h^0(C, L) - h^0(C, \omega_C \otimes L^{-1}) = \deg L + 1 - g.$$

Proof. See [Har77, IV, 1.3]. □

Lemma 4 (Line bundles on an elliptic curve)**.** *Let E be an elliptic curve. Then $\omega_E \cong \mathcal{O}_E$. If L is a line bundle with positive degree e, then $h^0(E, L) = e$. If L has degree zero and is non-trivial, then $h^0(E, L) = 0$.*

Proof. Applying the Riemann–Roch formula, we have

$$h^0(E, \mathcal{O}_E) - h^0(E, \omega_E) = \deg \mathcal{O}_E = 0,$$
$$h^0(E, \omega_E) - h^0(E, \mathcal{O}_E) = \deg \omega_E = 0.$$

We see that $H^0(E, \omega_E) \neq \{0\}$. A non-zero section of ω_E has an effective divisor of zeros with degree zero, so it will be nowhere vanishing on E. Hence its local equation gives a surjection $\omega_E \twoheadrightarrow \mathcal{O}_E$, so trivializes ω_E.

A line bundle of negative degree carries no non-trivial global sections. So the Riemann–Roch formula gives

$$h^0(E, L) = h^0(E, L) - h^0(E, \omega_E \otimes L^{-1}) = \deg L = e$$

when $e > 0$. If $\deg L = 0$, a non-zero section will trivialize L as for ω_E. □

Now, we fix an elliptic curve E, an ample Cartier divisor A of degree $d > 0$, and a Cartier divisor D such that $\mathcal{O}_E(D) \in \mathrm{Pic}^0(E)$ is non-torsion, i.e., it is not a trivial line bundle in any positive tensor power. Thus $\mathcal{O}_E(D)$ carries only trivial global sections.

Set

$$f : X = \mathbb{P}_E\big(\mathcal{O}_E \oplus \mathcal{O}_E(A)\big) \longrightarrow E,$$

and denote by ξ the divisor corresponding to $\mathcal{O}_X(1)$, up to linear equivalence. Note that X is a smooth surface. The two projections

$$\mathcal{O}_E \oplus \mathcal{O}_E(A) \longrightarrow \mathcal{O}_E, \qquad \mathcal{O}_E \oplus \mathcal{O}_E(A) \longrightarrow \mathcal{O}_E(A)$$

define two sections $s_0 : E \to X$ and $s_\infty : E \to X$. Note that $f : X \to E$ is a separated morphism, so its sections are closed immersions. Their images S_0 and S_∞ are closed subvarieties of X.

Lemma 5. *The images S_0 and S_∞ of the two sections are smooth, disjoint and isomorphic to E. Their classes and intersection numbers are*

$$S_0 \sim \xi - f^*A, \qquad S_\infty \sim \xi, \qquad \xi|_{S_0} \sim 0, \qquad \xi|_{S_\infty} \sim A,$$
$$\xi^2 = d, \qquad S_0^2 = -d, \qquad S_\infty^2 = d, \qquad S_0 \cdot S_\infty = 0.$$

Proof. For

$$f : X = \mathbb{P}_E\big(\mathcal{O}_E \oplus \mathcal{O}_E(A)\big) \longrightarrow E, \qquad \deg A = d,$$

the two projections

$$\mathcal{O}_E \oplus \mathcal{O}_E(A) \longrightarrow \mathcal{O}_E, \qquad \mathcal{O}_E \oplus \mathcal{O}_E(A) \longrightarrow \mathcal{O}_E(A)$$

define two sections $s_0, s_\infty : E \to X$ with images S_0, S_∞ as divisors. Take an open subset $U \subset E$ which trivializes $\mathcal{O}_E(A)$. Then

$$f^{-1}(U) \cong U \times \mathbb{P}^1,$$

where $\mathbb{P}^1$ has homogeneous coordinates $[x : y]$. The summand $\mathcal{O}_E$ corresponds to x, and $\mathcal{O}_E(A)$ corresponds to y, so

$$S_0 \cap f^{-1}(U) = V_+(y) = U \times \{[1:0]\},$$

$$S_\infty \cap f^{-1}(U) = V_+(x) = U \times \{[0:1]\}.$$

Thus they are disjoint. Since s_0, s_∞ are closed immersions, $S_0 \cong E \cong S_\infty$, and they are nonsingular.

Now the tautological quotient

$$\rho : f^*\big(\mathcal{O}_E \oplus \mathcal{O}_E(A)\big) = \mathcal{O}_X \oplus f^*\mathcal{O}_E(A) \twoheadrightarrow \mathcal{O}_X(1)$$

induces two homomorphisms

$$\alpha : \mathcal{O}_X \longrightarrow \mathcal{O}_X(1), \qquad \beta : f^*\mathcal{O}_E(A) \longrightarrow \mathcal{O}_X(1).$$

Locally, on an open subset $U = \operatorname{Spec} R \subset E$ which trivializes $\mathcal{O}_E(A)$, write

$$\mathcal{O}_E(A)|_U = \widetilde{R\langle e\rangle}.$$

Then

$$f^{-1}(U) = U \times \mathbb{P}^1 = \operatorname{Proj} R[x, y].$$

Choose the affine chart

$$D_+(x) \cong \operatorname{Spec} R[y/x].$$

For $f_1 : \operatorname{Spec} R[y/x] \to \operatorname{Spec} R$, we have

$$f^*\mathcal{O}_E|_{D_+(x)} = f_1^*\widetilde{R} = R \otimes_R \widetilde{R[y/x]} = \widetilde{R[y/x]},$$

$$f^*\mathcal{O}_E(A)|_{D_+(x)} = f_1^*\widetilde{R\langle e\rangle} = R\langle e\rangle \otimes_R \widetilde{R[y/x]} = \widetilde{R[y/x]\langle e\rangle},$$

and

$$\mathcal{O}_X(1)|_{D_+(x)} = \widetilde{R[y/x]\langle x\rangle}.$$

For $f_2 : \operatorname{Spec} R[x/y] \to \operatorname{Spec} R$, replace y/x by x/y.

Therefore, locally ρ is the homomorphism

$$\begin{aligned} \widetilde{R[y/x]} \oplus \widetilde{R[y/x]\langle e\rangle} &\longrightarrow \widetilde{R[y/x]\langle x\rangle}, \\ (1, 0) &\longmapsto x, \\ (0, e) &\longmapsto y = (y/x)x. \end{aligned}$$

Thus β locally sends the generator e to $y = (y/x)x$, that is,

$$\widetilde{R[y/x]\langle e\rangle} \xrightarrow{\cong} \widetilde{R[y/x]\langle x\rangle} \xrightarrow{\cdot(y/x)} \widetilde{R[y/x]\langle x\rangle}.$$

On the other hand, since

$$\operatorname{Hom}\big(f^*\mathcal{O}_E(A), \mathcal{O}_X(1)\big) \cong H^0\big(X, f^*\mathcal{O}_E(-A) \otimes \mathcal{O}_X(1)\big),$$

we may regard β as a global section on X, whose divisor of zeros is locally defined by $y/x = 0$. On $x \neq 0$, its vanishing order is one, corresponding to S_0, and on $y \neq 0$ it is nonvanishing. Hence

$$\mathcal{O}_X(S_0) \cong f^*\mathcal{O}_E(-A) \otimes \mathcal{O}_X(1),$$

which implies $S_0 \sim \xi - f^*A$.

The homomorphism α locally sends 1 to x, and corresponds to a global section of $\mathcal{O}_X(1)$ on X. On $y \neq 0$, its vanishing order is one, corresponding to S_∞, and on $x \neq 0$ it is nonvanishing. Thus $S_\infty \sim \xi$.

On S_0, x is nowhere vanishing, so α induces an isomorphism

$$\mathcal{O}_{S_0} \xrightarrow{\cdot x} \mathcal{O}_X(1)|_{S_0}.$$

On S_∞, y is nowhere vanishing, so β induces an isomorphism

$$(f|_{S_\infty})^*\mathcal{O}_E(A) \xrightarrow{\beta|_{S_\infty}} \mathcal{O}_X(1)|_{S_\infty}.$$

Via $f|_{S_i} : S_i \cong E$, these give

$$\xi|_{S_0} \sim 0, \qquad \xi|_{S_\infty} \sim A.$$

Moreover,

$$\xi^2 = S_\infty^2 = \deg_{S_\infty}\big(\mathcal{O}_X(1)|_{S_\infty}\big) = \deg_E A = d,$$

and

$$\begin{aligned} S_0^2 &= \deg_{S_0}\big(\mathcal{O}_X(1)|_{S_0} \otimes (f|_{S_0})^*\mathcal{O}_E(-A)\big) \\ &= \deg_{S_0}\big(\mathcal{O}_{S_0} \otimes \mathcal{O}_{S_0}(-A)\big) \\ &= \deg_E \mathcal{O}_E(-A) \\ &= -d. \end{aligned}$$

Finally, $S_0 \cdot S_\infty = 0$ because they are disjoint. □

Proposition 6. *In the situation above, the Cartier divisor $L = \xi + f^*D$ is nef and big. More precisely, for every integer $m > 0$,*

$$h^0(X, mL) = \frac{dm(m+1)}{2}, \qquad \kappa_\iota(X, L) = \nu(X, L) = 2,$$

where ν is the numerical dimension defined in [b-Div].

Proof. Since $\deg D = 0$, we see that $L \equiv \xi$. It is a nef Cartier divisor because $\xi \sim S_\infty$, where S_∞ is an irreducible effective curve and $S_\infty^2 = d > 0$. Hence $\nu(X, L) = 2$.

By Proposition 2, we have

$$\begin{aligned} f_*\mathcal{O}_X(mL) &\cong f_*\big(\mathcal{O}_X(m) \otimes f^*\mathcal{O}_E(mD)\big) \\ &\cong \mathrm{Sym}^m\big(\mathcal{O}_E \oplus \mathcal{O}_E(A)\big) \otimes \mathcal{O}_E(mD) \\ &\cong \bigoplus_{k=0}^{m} \mathcal{O}_E(kA + mD), \end{aligned}$$

where the third isomorphism comes from

$$\binom{m+1}{m} = m+1, \qquad \mathrm{Sym}^m\big(\mathcal{O}_E \oplus \mathcal{O}_E(A)\big) \cong \bigoplus_{k=0}^{m} \mathcal{O}_E(kA).$$

The summand with $k = 0$ has no non-zero sections because mD is non-trivial of degree zero. For $k > 0$, Lemma 4 gives $h^0(E, kA + mD) = kd$. Therefore,

$$h^0\big(X, \mathcal{O}_X(mL)\big) = \sum_{k=1}^{m} kd = \frac{dm(m+1)}{2} = \frac{d}{2}m^2 + O(m).$$

Thus $\kappa(X, L) = 2 = \dim X$, which implies that L is big. Since L is Cartier,

$$\kappa_\iota(X, L) = \kappa(X, L) = \nu(X, L) = 2.$$

For the asymptotic definition of the Iitaka dimension, see [BGR, Remark 3.1.37]. □

Recall that, by Zariski's Main Theorem, if $f : Y \to X$ is a proper birational morphism between normal algebraic varieties and L is a Cartier divisor on X, then L is semiample if and only if f^*L is, because pullback gives

$$H^0\big(X, \mathcal{O}_X(mL)\big) \cong H^0\big(Y, \mathcal{O}_Y(mf^*L)\big).$$

Proposition 7. *For the above X and L, the following hold.*

(1) *L is abundant but not log abundant with respect to (X, S_0). Its Cartier closure $\overline{L}$ is b-abundant but not b-log abundant with respect to this pair.*

(2) *L is log abundant with respect to the klt pair $(X, 0)$, but is not semiample. Consequently, $\overline{L}$ is b-log abundant with respect to $(X, 0)$ but is not b-semiample.*

Proof. (1) The pair (X, S_0) is log smooth and has lc singularities, with S_0 as its only lc centre; see [b-Div, Proposition 4.1.7]. Lemma 5 and $f|_{S_0} : S_0 \cong E$ give

$$L|_{S_0} = \xi|_{S_0} + (f^*D)|_{S_0} \sim D, \qquad \kappa_\iota(S_0, L|_{S_0}) = -\infty, \qquad \nu(S_0, L|_{S_0}) = 0.$$

Thus the restriction $L|_{S_0}$ is not abundant, whereas L is abundant by Proposition 6. Hence $\overline{L}$ is b-abundant, as it descends to X.

For any higher crepant log resolution $g : V \to X$, the strict transform S_0' of S_0 remains a coefficient-one component of the crepant boundary, by the discrepancy formula, and hence is an lc centre. The induced morphism $r : S_0' \to S_0$ is proper and birational, and

$$(\mathrm{tr}_V\, \overline{L})\big|_{S_0'} = (g^*L)|_{S_0'} = r^*(L|_{S_0}) \sim r^*D,$$

where the right-hand side is nef but not abundant, by [b-Div, Proposition 2.5.8]. That is, $\overline{L}$ is not b-log abundant with respect to (X, S_0).

(2) Since X is a smooth ruled surface, the pair $(X, 0)$ is klt and has no lc centres. Thus abundance of L on X implies that L is log abundant and $\overline{L}$ is b-log abundant with respect to $(X, 0)$.

But, for every $m > 0$, the divisor mD has degree zero and $\mathcal{O}_E(mD)$ is non-trivial. Hence every global section of $\mathcal{O}_X(mL)$ restricts to zero on S_0, so every point of S_0 is a base point of $|mL|$ for every $m > 0$. Thus L is not semiample, and, by the pullback observation above, $\overline{L}$ is not b-semiample. □

Remark 8 (Associated to the canonical bundle formula). By Proposition 2, we have

$$K_X \sim -2\xi + f^*(K_E + A) \sim -2\xi + f^*A.$$

By Lemma 5 and the definition of L, this gives

$$K_X + S_0 + L \sim f^*D.$$

Since $\overline{L}$ descends to X and (X, S_0) is log smooth, $(X, S_0, \overline{L})$ is g-lc. The identity of X is a log resolution and $\lceil \mathbf{A}^*(X, S_0, \overline{L})_X \rceil = 0$. Since $f_*\mathcal{O}_X = \mathcal{O}_E$, the rank-one condition holds and $f : (X, S_0, \overline{L}) \to E$ is a g-lc-trivial fibration.

For each closed point $p \in E$, let $F_p = f^*p$. It is a smooth curve and meets S_0 transversely at one point, so $(X, S_0 + F_p)$ is log smooth and has lc singularities. Consider

$$t_p = \sup\big\{ s \in \mathbb{R} \mid (X, S_0 + sF_p) \text{ is sub-lc over } p \big\}.$$

Since $\overline{L}$ descends to X, this is also the threshold for the g-pair. The snc condition gives $t_p = 1$. Thus the discriminant b-divisor $\mathbf{C}$ satisfies

$$\operatorname{mult}_p C_E = 1 - t_p = 0$$

for every closed point $p \in E$, so $C_E = 0$. For the moduli b-divisor $\mathbf{N}$ defined using D,

$$K_E + C_E + N_E = D.$$

Taking $K_E = 0$, we get $N_E = D$. As in the proof of Proposition 7, D is nef of degree zero and is not abundant. So the moduli b-divisor $\mathbf{N}$ is not b-abundant.

References

[Har77] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer-Verlag, 1977.

[BGR] *Birational Geometry Seminar: Chapters 1–5 and Appendices A–D. Revised Reading Notes.*

[b-Div] *b-Divisors and Abundance for Generalized Pairs.*